

Fake News Detection — Project Report

Bidirectional LSTM Text Classifier | Model Development & Evaluation Summary

1. Objective & Approach

The goal of this project is to classify news articles as **Real** or **Fake** based purely on the article text, using a deep learning text classifier. The dataset (**news_dataset.csv**) contains 5,829 labeled articles (2,834 fake, 2,995 real). Text was tokenized and converted to padded integer sequences (vocabulary size 20,000, max sequence length 300 tokens) and fed into a Bidirectional LSTM network. The labeled data was split 80/20 into training (4,663 articles) and test (1,166 articles) sets using a random split, since articles are not date-ordered and there is no forward-in-time prediction requirement for this task.

2. Data Quality & Leakage Findings

Before trusting the headline accuracy figure, the dataset was audited for shortcuts the model could exploit instead of learning genuine linguistic patterns of misinformation:

- **Wire-service leakage (critical).** 99.6% of articles labeled *Real* contain the word "Reuters" in the text, versus only 1.8% of articles labeled *Fake*. This means the model can reach very high accuracy simply by detecting wire-service boilerplate (e.g. a dateline like "WASHINGTON (Reuters) -") rather than learning to distinguish misinformation from legitimate reporting on content or style grounds. This is a well-known issue with this public dataset and is the most likely explanation for the near-perfect scores reported below.
- **Stylistic confounds.** Fake-labeled articles disproportionately use clickbait formatting — ALL-CAPS phrases, exclamation marks, and rhetorical framing (e.g. "WATCH:", "WSJ REPORTER RIPS INTO...") — which is a legitimate signal, but it is style rather than a semantic fact-check, so the model does not verify factual accuracy.
- **No missing values or malformed rows** were found in the text or label columns.
- **Class balance** is reasonable (48.6% fake / 51.4% real), so no resampling was required.

3. Model Architecture

Layer	Output Shape	Parameters
Embedding (vocab=20,000, dim=128)	(None, 300, 128)	2,560,000
Bidirectional LSTM (64 units)	(None, 128)	98,816
Dropout (0.5)	(None, 128)	0
Dense (32, ReLU)	(None, 32)	4,128
Dropout (0.3)	(None, 32)	0
Dense (1, Sigmoid)	(None, 1)	33

Total parameters: 2,662,977 (all trainable, ~10.16 MB). Optimizer: Adam. Loss: Binary Crossentropy. Training used early stopping (patience = 3, monitored on validation loss, best weights restored) and ran for 5 epochs before stopping.

4. Validation Results

Metric	Train	Test
Accuracy	100.0%	99.66%
Precision	—	99.50%
Recall	—	99.83%

Evaluated on the 1,166-article held-out test split (562 fake, 601 real actual counts per the confusion matrix below). Given the wire-service leakage identified in Section 2, these numbers should be read as an upper bound achieved partly through a formatting shortcut, not solely through content-based fake-news detection. See Section 6 for recommendations to obtain a more trustworthy estimate.

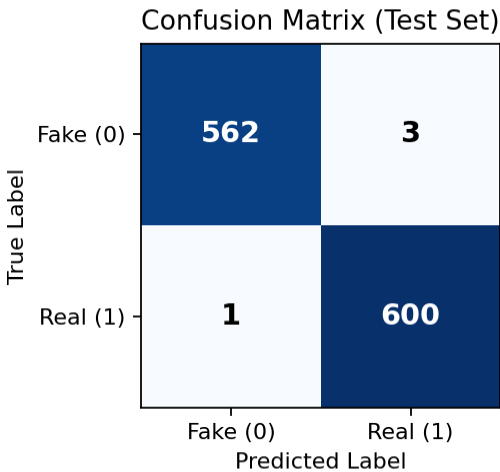


Figure 1 — Confusion matrix on the test set.

5. Training Curves

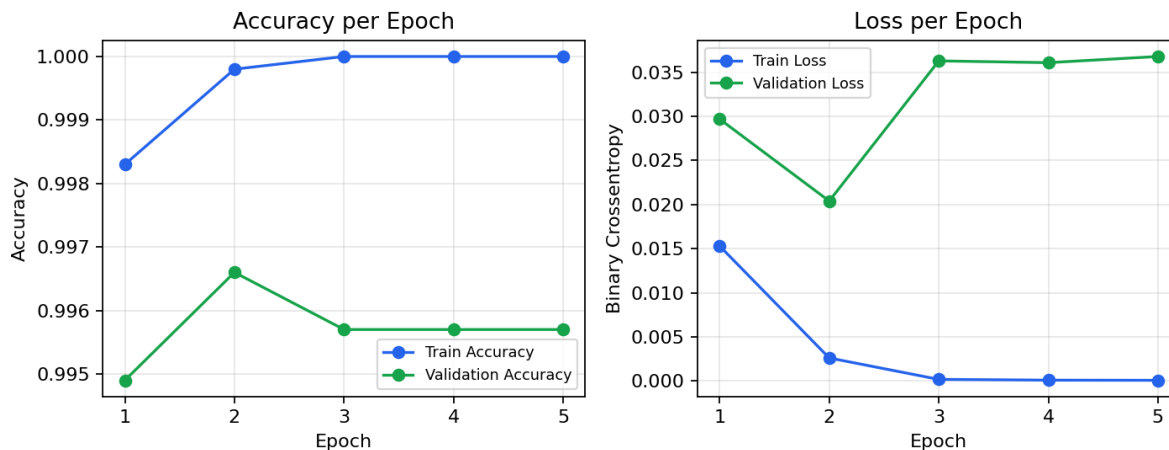


Figure 2 — Accuracy and loss per epoch. Training accuracy reaches 100% by epoch 3 while training loss keeps falling, but validation loss bottoms out at epoch 2 (0.0204) and rises afterward — a classic overfitting signature. Early stopping correctly restored the epoch-2 weights as the best checkpoint.

6. Limitations & Recommendations

- **Remove the leakage source before deployment.** Strip wire-service datelines/agency names (e.g. "Reuters") from the text during preprocessing, then retrain — this gives a realistic estimate of how well the model detects fake news from content and writing style alone, rather than source formatting.
- **Test on out-of-distribution articles.** This dataset covers a specific time period and set of publishers; accuracy should be re-validated on newer or differently-sourced articles before trusting it in production, the same way a time-based holdout is needed for genuinely forward-looking predictions.
- **Add regularization or reduce capacity** if retraining without the leakage signal, since a smaller, content-driven signal is likely to be noisier and more prone to overfitting than the current shortcut.
- **Track precision/recall separately per class** in production, since the cost of mislabeling real news as fake (censorship risk) differs from missing actual fake news (misinformation risk).

7. Project Files

File	Purpose
news_dataset.csv	Labeled training data (article text + 0/1 label)
News_Fake_Detection.ipynb	Full training / evaluation notebook
model.pkl	Trained Keras BiLSTM classifier (pickled)
Tokenizer.pkl	Fitted Keras Tokenizer (vocab=20,000) used to convert text to sequences
app.py	Streamlit web app for interactive predictions